

MF27 Formula List

List of Formulae and Results · Singapore-Cambridge A-Level · used from 2025

H1 Mathematics (8865), H2 Mathematics (9758), H3 Mathematics and H2 Further Mathematics (9649)

Algebraic series

H2 Maths

Binomial expansions and Maclaurin series — Sequences & Series and the small- x approximations.

Binomial expansion, n a positive integer

$$(a + b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \binom{n}{3}a^{n-3}b^3 + \dots + b^n$$

where

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

Maclaurin series

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \dots + \frac{x^n}{n!}f^{(n)}(0) + \dots$$

Binomial series, any n

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \dots + \frac{n(n-1)\dots(n-r+1)}{r!}x^r + \dots$$

valid for $|x| < 1$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^r}{r!} + \dots$$

valid for all x

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots + \frac{(-1)^r x^{2r+1}}{(2r+1)!} + \dots$$

valid for all x

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots + \frac{(-1)^r x^{2r}}{(2r)!} + \dots$$

valid for all x

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots + \frac{(-1)^{r+1} x^r}{r} + \dots$$

valid for $-1 < x \leq 1$

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AP: nth term and sum

$$u_n = a + (n - 1)d, \quad S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + l)$$

GP: nth term and sum

$$u_n = ar^{n-1}, \quad S_n = \frac{a(1 - r^n)}{1 - r}$$

GP: sum to infinity

$$S_\infty = \frac{a}{1 - r}$$

valid for $|r| < 1$

Sum and nth term

$$u_n = S_n - S_{n-1}$$

Small-angle approximations (x in radians)

$$\sin x \approx x, \quad \cos x \approx 1 - \frac{x^2}{2}, \quad \tan x \approx x$$

Expanding $(a + bx)^n$: take out a^n first

$$(a + bx)^n = a^n \left(1 + \frac{b}{a}x\right)^n$$

valid for $\left|\frac{b}{a}x\right| < 1$

Partial fractions

H2 Maths

Splitting a proper fraction before you integrate it or expand it as a series.

Distinct linear factors

$$\frac{px + q}{(ax + b)(cx + d)} = \frac{A}{ax + b} + \frac{B}{cx + d}$$

A repeated linear factor

$$\frac{px^2 + qx + r}{(ax + b)(cx + d)^2} = \frac{A}{ax + b} + \frac{B}{cx + d} + \frac{C}{(cx + d)^2}$$

A quadratic factor that does not factorise

$$\frac{px^2 + qx + r}{(ax + b)(x^2 + c^2)} = \frac{A}{ax + b} + \frac{Bx + C}{x^2 + c^2}$$

NOT ON MF27 — MEMORISEImproper fraction (top degree $\geq$ bottom degree): divide first

$$\frac{x^2 + 1}{(x - 1)(x + 2)} = 1 + \frac{-x + 3}{(x - 1)(x + 2)}$$

Trigonometry

H2 Maths

Compound and double angles, and the principal values of the inverse functions. The factor formulae are gone from MF27.

$$\sin(A \pm B) \equiv \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) \equiv \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) \equiv \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin 2A \equiv 2 \sin A \cos A$$

$$\cos 2A \equiv \cos^2 A - \sin^2 A$$

$$\equiv 2 \cos^2 A - 1$$

$$\equiv 1 - 2 \sin^2 A$$

$$\tan 2A \equiv \frac{2 \tan A}{1 - \tan^2 A}$$

Principal values:

$$-\frac{1}{2}\pi \leq \sin^{-1} x \leq \frac{1}{2}\pi$$

valid for $|x| \leq 1$

$$0 \leq \cos^{-1} x \leq \pi$$

valid for $|x| \leq 1$

$$-\frac{1}{2}\pi < \tan^{-1} x < \frac{1}{2}\pi$$

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Pythagorean identities

$$\sin^2 A + \cos^2 A \equiv 1, \quad 1 + \tan^2 A \equiv \sec^2 A, \quad 1 + \cot^2 A \equiv \operatorname{cosec}^2 A$$

R-formula

$$a \sin \theta + b \cos \theta \equiv R \sin(\theta + \alpha), \quad R = \sqrt{a^2 + b^2}, \quad \tan \alpha = \frac{b}{a}$$

cos 2A turned round (for integrating $\sin^2$ and $\cos^2$)

$$\sin^2 A \equiv \frac{1}{2}(1 - \cos 2A), \quad \cos^2 A \equiv \frac{1}{2}(1 + \cos 2A)$$

Factor formulae (were on MF26, not on MF27)

$$\sin P + \sin Q \equiv 2 \sin \frac{1}{2}(P + Q) \cos \frac{1}{2}(P - Q)$$

$$\sin P - \sin Q \equiv 2 \cos \frac{1}{2}(P + Q) \sin \frac{1}{2}(P - Q)$$

$$\cos P + \cos Q \equiv 2 \cos \frac{1}{2}(P + Q) \cos \frac{1}{2}(P - Q)$$

$$\cos P - \cos Q \equiv -2 \sin \frac{1}{2}(P + Q) \sin \frac{1}{2}(P - Q)$$

Derivatives

H2 Maths

Only the five awkward derivatives are given. Everything else in differentiation is yours to remember.

$f(x)$	$f'(x)$
$\sin^{-1} x$	$\frac{1}{\sqrt{1-x^2}}$
$\cos^{-1} x$	$-\frac{1}{\sqrt{1-x^2}}$
$\tan^{-1} x$	$\frac{1}{1+x^2}$
$\operatorname{cosec} x$	$-\operatorname{cosec} x \cot x$
$\sec x$	$\sec x \tan x$

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Basic derivatives

$$\frac{d}{dx} \sin x = \cos x, \quad \frac{d}{dx} \cos x = -\sin x, \quad \frac{d}{dx} \tan x = \sec^2 x, \quad \frac{d}{dx} \cot x = -\operatorname{cosec}^2 x$$

$$\frac{d}{dx} e^x = e^x, \quad \frac{d}{dx} \ln x = \frac{1}{x}, \quad \frac{d}{dx} a^x = a^x \ln a$$

Product and quotient rules

$$\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}, \quad \frac{d}{dx} \left(\frac{u}{v} \right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

Chain rule and parametric form

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}, \quad \frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$

Integrals

H2 Maths

The eight standard forms. Constants of integration are left out and a is a positive constant.

$f(x)$	$\int f(x) dx$	Valid for
$\frac{1}{x^2 + a^2}$	$\frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right)$	
$\frac{1}{\sqrt{a^2 - x^2}}$	$\sin^{-1} \left(\frac{x}{a} \right)$	$ x < a$
$\frac{1}{x^2 - a^2}$	$\frac{1}{2a} \ln \left(\frac{x - a}{x + a} \right)$	$x > a$
$\frac{1}{a^2 - x^2}$	$\frac{1}{2a} \ln \left(\frac{a + x}{a - x} \right)$	$ x < a$
$\tan x$	$\ln(\sec x)$	$ x < \frac{1}{2}\pi$
$\cot x$	$\ln(\sin x)$	$0 < x < \pi$
$\operatorname{cosec} x$	$-\ln(\operatorname{cosec} x + \cot x)$	$0 < x < \pi$
$\sec x$	$\ln(\sec x + \tan x)$	$ x < \frac{1}{2}\pi$

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Basic integrals

$$\int x^n dx = \frac{x^{n+1}}{n+1} \quad (n \neq -1), \quad \int \frac{1}{x} dx = \ln|x|, \quad \int e^{ax} dx = \frac{1}{a} e^{ax}$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad \int \cos ax dx = \frac{1}{a} \sin ax, \quad \int \sec^2 ax dx = \frac{1}{a} \tan ax$$

Standard forms

$$\int \frac{f'(x)}{f(x)} dx = \ln|f(x)|, \quad \int [f(x)]^n f'(x) dx = \frac{[f(x)]^{n+1}}{n+1}$$

Integration by parts

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

Volume of revolution

$$V = \pi \int_a^b y^2 dx \quad (\text{about the } x\text{-axis}), \quad V = \pi \int_c^d x^2 dy \quad (\text{about the } y\text{-axis})$$

Vectors

H2 Maths

The ratio theorem and the cross product. Lines, planes, angles and distances are not given.

The point dividing AB in the ratio $\lambda : \mu$

$$\frac{\mu \mathbf{a} + \lambda \mathbf{b}}{\lambda + \mu}$$

Vector (cross) product

$$\mathbf{a} \times \mathbf{b} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \times \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = \begin{pmatrix} a_2 b_3 - a_3 b_2 \\ a_3 b_1 - a_1 b_3 \\ a_1 b_2 - a_2 b_1 \end{pmatrix}$$

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Scalar product

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos \theta = a_1 b_1 + a_2 b_2 + a_3 b_3$$

Length of projection of a on b, and area of triangle

$$|\mathbf{a} \cdot \hat{\mathbf{b}}|, \quad \text{Area} = \frac{1}{2} |\mathbf{a} \times \mathbf{b}|$$

Line and plane

$$\mathbf{r} = \mathbf{a} + \lambda \mathbf{d}, \quad \mathbf{r} \cdot \mathbf{n} = D$$

Distance from point P to the plane $\mathbf{r} \cdot \mathbf{n} = D$

$$\frac{|\mathbf{p} \cdot \mathbf{n} - D|}{|\mathbf{n}|}$$

Angle between line and plane, and between two planes

$$\sin \theta = \frac{|\mathbf{d} \cdot \mathbf{n}|}{|\mathbf{d}| |\mathbf{n}|}, \quad \cos \theta = \frac{|\mathbf{n}_1 \cdot \mathbf{n}_2|}{|\mathbf{n}_1| |\mathbf{n}_2|}$$

Applications of definite integrals

Further Maths

Further Maths: the length of a curve and the area of the surface it sweeps out. New in MF27.

Arc length, y as a function of x

$$s = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

Surface area of revolution about the x-axis

$$S = \int_a^b 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

Functions of two variables

Further Maths

Further Maths: approximating $f(x, y)$ near a point (a, b) , as Maclaurin does for one variable. New in MF27.

Quadratic approximation of f at (a, b)

$$\begin{aligned} f(x, y) \approx & f(a, b) \\ & + f_x(a, b)(x - a) + f_y(a, b)(y - b) \\ & + \frac{1}{2} f_{xx}(a, b)(x - a)^2 \\ & + f_{xy}(a, b)(x - a)(y - b) \\ & + \frac{1}{2} f_{yy}(a, b)(y - b)^2 \end{aligned}$$

Numerical methods

Further Maths

Further Maths: estimating an integral, a root, or a step of a differential equation. Not in H2 Maths.

Trapezium rule, one strip

$$\int_a^b f(x) dx \approx \frac{1}{2}(b-a)[f(a) + f(b)]$$

Simpson's rule, two strips

$$\int_a^b f(x) dx \approx \frac{1}{6}(b-a) \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right]$$

Newton-Raphson, x_1 a first approximation to a root of $f(x) = 0$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

Euler method, step size h

$$y_2 = y_1 + hf(x_1, y_1)$$

Improved Euler method, step size h

$$u_2 = y_1 + hf(x_1, y_1), \quad y_2 = y_1 + \frac{h}{2}[f(x_1, y_1) + f(x_2, u_2)]$$

Standard distributions

H1 & H2 Maths

H1 and H2 Maths use the binomial row only. Poisson, geometric and exponential are Further Maths.

DISCRETE

Distribution of X	$P(X = x)$	Mean	Variance
Binomial B(n, p)	$\binom{n}{x} p^x (1-p)^{n-x}$	np	$np(1-p)$
Poisson Po(λ)	$e^{-\lambda} \frac{\lambda^x}{x!}$	λ	λ
Geometric Geo(p)	$(1-p)^{x-1} p$	$\frac{1}{p}$	$\frac{1-p}{p^2}$

CONTINUOUS

Distribution of X	p.d.f.	Mean	Variance
Exponential	$\lambda e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$

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Expectation and variance

$$E(X) = \sum x P(X = x), \quad \text{Var}(X) = E(X^2) - [E(X)]^2$$

Linear combinations (X, Y independent for the variance)

$$E(aX + b) = aE(X) + b, \quad \text{Var}(aX \pm bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y)$$

Sample mean (central limit theorem, n large)

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \text{ approximately}$$

Probability

$$P(A \cup B) = P(A) + P(B) - P(A \cap B), \quad P(A | B) = \frac{P(A \cap B)}{P(B)}$$

Independent events, and counting

$$P(A \cap B) = P(A)P(B), \quad {}^n P_r = \frac{n!}{(n-r)!}, \quad {}^n C_r = \frac{n!}{r!(n-r)!}$$

Sampling and testing

H1 & H2 Maths

The unbiased estimate of the population variance from a sample. MF27 no longer gives the two-sample pooled version.

Unbiased estimate of population variance

$$s^2 = \frac{n}{n-1} \left(\frac{\sum (x - \bar{x})^2}{n} \right) \\ = \frac{1}{n-1} \left(\sum x^2 - \frac{(\sum x)^2}{n} \right)$$

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Unbiased estimate of the mean

$$\bar{x} = \frac{\sum x}{n}$$

z-test statistic for a mean

$$Z = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$$

Regression and correlation

H1 & H2 Maths

r and the y-on-x line. Your GC gives both; the formula is for questions that hand you summary totals.

Product moment correlation coefficient

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}} \\ = \frac{\sum xy - \frac{\sum x \sum y}{n}}{\sqrt{\left(\sum x^2 - \frac{(\sum x)^2}{n} \right) \left(\sum y^2 - \frac{(\sum y)^2}{n} \right)}}$$

Regression line of y on x

$$y - \bar{y} = b(x - \bar{x}), \quad b = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$

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Regression line of x on y (not given — swap the roles)

$$x - \bar{x} = d(y - \bar{y}), \quad d = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (y - \bar{y})^2}$$

Both lines pass through the mean point

$$(\bar{x}, \bar{y})$$

Wilcoxon signed rank test

Further Maths

Further Maths non-parametric testing. Reject the null hypothesis when T is at most the value in the table.

P = sum of the ranks of the positive differences.

Q = sum of the ranks of the negative differences.

T = the smaller of P and Q.

Each entry is the LARGEST T that still rejects the null hypothesis at that level.

A dash: no T can reject at that level for that n.

CRITICAL VALUES OF T

n	One-tail 0.05 · Two-tail 0.1	One-tail 0.025 · Two-tail 0.05	One-tail 0.01 · Two-tail 0.02	One-tail 0.005 · Two-tail 0.01
6	2	0	—	—
7	3	2	0	—
8	5	3	1	0
9	8	5	3	1
10	10	8	5	3
11	13	10	7	5
12	17	13	9	7
13	21	17	12	9
14	25	21	15	12
15	30	25	19	15
16	35	29	23	19
17	41	34	27	23
18	47	40	32	27
19	53	46	37	32
20	60	52	43	37

Mathematical results

Further Maths & H3

Inequalities and counting for proofs in Further Maths and H3. A new page in MF27.

AM-GM inequality — $x_1, \dots, x_n \geq 0$; equal only when all the x_i are equal

$$\frac{x_1 + x_2 + \dots + x_n}{n} \geq \sqrt[n]{x_1 x_2 \dots x_n}$$

Cauchy-Schwarz inequality — equal only when $u_i = kv_i$ for every i , for some $k \neq 0$

$$\left(\sum_{i=1}^n u_i v_i \right)^2 \leq \left(\sum_{i=1}^n u_i^2 \right) \left(\sum_{i=1}^n v_i^2 \right)$$

Triangle inequality — equal when the x_i all have the same sign (or are zero)

$$|x_1 + x_2 + \dots + x_n| \leq |x_1| + |x_2| + \dots + |x_n|$$

Inclusion-exclusion principle

$$\begin{aligned} |A_1 \cup \dots \cup A_n| &= \sum_i |A_i| - \sum_{i < j} |A_i \cap A_j| \\ &\quad + \sum_{i < j < k} |A_i \cap A_j \cap A_k| \\ &\quad - \dots \\ &\quad + (-1)^{n-1} |A_1 \cap \dots \cap A_n| \end{aligned}$$

Complex numbers (not on the list at all)

H2 Maths

A whole H2 topic with nothing given. Since 2025 H2 Maths uses Cartesian form only — no polar or exponential form.

Modulus and conjugate of $z = x + iy$

$$|z| = \sqrt{x^2 + y^2}, \quad z^* = x - iy, \quad zz^* = |z|^2$$

Real and imaginary parts

$$z + z^* = 2\operatorname{Re}(z), \quad z - z^* = 2i\operatorname{Im}(z)$$

Dividing: multiply top and bottom by the conjugate

$$\frac{1}{z} = \frac{z^*}{|z|^2}$$

Argument: $\tan$ of the angle, then check the quadrant on the Argand diagram

$$\tan(\arg z) = \frac{y}{x}, \quad -\pi < \arg z \leq \pi$$

A polynomial with real coefficients

$$p(z) = 0 \implies p(z^*) = 0$$

Typeset by Adrian's Math Tuition from the list SEAB publishes. The yellow boxes are formulas the list does NOT give. In the exam, the official booklet is the one handed out. Online: adrianmathtuition.com/formulas/mf27